Quantitative Diagrammatic Reasoning for Probabilistic Processes

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joint work with Fabio Zanasi

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Outline

Algebraic Semantics, Briefly

Idea: programming syntax as algebraic operations (pioneered in [SS71] and [Gog+77]).

$$;(P, Q) = P \,;\, Q \qquad\qquad \mathsf{ifte}(P, Q, Q') = \mathsf{if}\ P\ \mathsf{then}\ Q\ \mathsf{else}\ Q'$$

This enables equational reasoning for program equivalence:

$$\dfrac{\dfrac{P \,;\, (Q \,;\, R) = (P \,;\, Q) \,;\, R \qquad\qquad \mathsf{ifte}(P, Q, Q) = P \,;\, Q}{\vdots}}{\mathsf{ifte}(P, Q, Q) \,;\, R = \mathsf{ifte}(P, Q \,;\, R, Q \,;\, R)}$$

...among other things.

[SS71] Dana Scott and Christopher Strachey. "Toward a Mathematical Semantics for Computer Languages". In: *Proceedings of the Symposium on Computers and Automata*. Vol. 21. 1971

[Gog+77] J. A. Goguen et al. "Initial Algebra Semantics and Continuous Algebras". In: *J. ACM* 24.1 (1977), 68–95. ISSN: 0004-5411. DOI: 10.1145/321992.321997. URL: https://doi.org/10.1145/321992.321997

Problems with Randomness 1: Approximations

Example ([Neu51])

```
return flip(0.5)
```

```
do
    x = flip(p)
    y = flip(p)
while (x == y)
return x
```

As long as the bias is consistent and not total ($0\% < p < 100\%$), the two programs have the same behavior.

[Neu51] John von Neumann. “Various Techniques Used in Connection with Random Digits”. In: *National Bureau of Standards Applied Math Series*. Ed. by G. E. Forsythe. Vol. 12. Reprinted in von Neumann’s Collected Works, Volume 5, Pergamon Press, 1963, pp. 768–770. National Bureau of Standards, 1951, pp. 36–38

Problems with Randomness 1: Approximations

Example (Guaranteed Termination)

```
return flip(0.5)
```

```
i = 0
do
    i = i + 1
    x = flip(p)
    y = flip(p)
while (x == y) AND i <= 1000
return x
```

The second program is very close to being a fair coin flip. We need a finer comparison than exact equivalence.

Problems with Randomness 2: Substitution

Example (Reusing results)

```
x = flip(0.5)
return x or x
```

```
return flip(0.5) or flip(0.5)
```

The semantics of these two programs are two *different* distributions:

$$\begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix} \qquad\qquad \begin{bmatrix} 0.75 \\ 0.25 \end{bmatrix}$$

We need a *resource sensitive* language to reason about these programs.

Overview

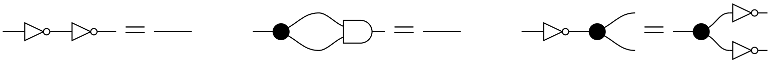
Our solution is quantitative diagrammatic reasoning. It combines quantitative equations and string diagrams.

Quantitative Equations.

- Replace exact equivalence with a distance
- Reason equationally: $P =_\varepsilon Q$ means P and Q are at distance $\leq \varepsilon$.

String Diagrams.

- Processes are morphisms in monoidal categories
- String diagrams represent such morphisms
- Reason equationally:



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$\mathcal{V}$-Relations

We define a kind of fuzzy/many-valued equality.

$$\frac{= \subseteq X \times X}{X \times X \to \{0,1\}} *$$

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The **quantities** in $\mathcal{V}$ form a quantale: $(\mathcal{V}, \sqsubseteq, \oplus)$, e.g. $(\{0,1\}, \vdash, \vee)$ or $([0,\infty], \geq, +)$.

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Definition

- A **$\mathcal{V}$-relation** is a function $d : X \times X \to \mathcal{V}$.
- A morphism $f : (X, d_X) \to (Y, d_Y)$ is a function $f : X \to Y$ satisfying $d_X(x,y) \sqsubseteq d_Y(f(x), f(y))$ for all $x, y \in X$.
- $\mathcal{V}$**Rel** is the category of $\mathcal{V}$-relations.

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Quantitative Equations

Proposition

A $\mathcal{V}$-relation (X, d) is determined by the relations $=_\varepsilon \subseteq X \times X$.

$$d(x, y) \sqsupseteq \varepsilon \quad \Longleftrightarrow \quad x =_\varepsilon y$$

Quantitative Equations

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$$\begin{array}{rr} \text{if } \varepsilon \sqsupseteq \varepsilon' \text{ and } x =_\varepsilon y\text{, then } x =_{\varepsilon'} y & \textit{(monotone)} \\ \text{if } x =_{\varepsilon_i} y \text{ for all } i \in I\text{, then } x =_{\sup_i \varepsilon_i} y & \textit{(continuous)} \end{array} \qquad (*)$$

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$$\frac{\varepsilon \sqsupseteq \varepsilon' \qquad x =_\varepsilon y}{x =_{\varepsilon'} y}\,\textsc{Mon} \qquad\qquad \frac{\forall i \in I,\ x =_{\varepsilon_i} y}{x =_{\sup_i \varepsilon_i} y}\,\textsc{Cont} \tag{$*$}$$

This enables quantitative equational reasoning to study $\mathcal{V}$-relations.

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Example

$$\frac{}{x =_\top x} \qquad \frac{x =_\top y}{x = y} \qquad \frac{x =_\varepsilon y}{y =_\varepsilon x} \qquad \frac{x =_\varepsilon y \qquad y =_\delta z}{x =_{\varepsilon \oplus \delta} z} \qquad \frac{x =_\top y \qquad y =_\top x}{x = y}$$

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It gets more interesting when X carries more structure.

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Given (X, d_X) and (Y, d_Y), there is a $\mathcal{V}$-relation on $X \times Y$ defined by

$$d((x,y),(x',y')) = d_X(x,x') \oplus d_Y(y,y').$$

If $\oplus$ is commutative, we get a symmetric monoidal category (SMC) $(\mathcal{V}\mathbf{Rel}, \boxtimes_{\oplus}, \mathbf{1})$.

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A $\mathcal{V}$**Rel-category** is a category $\mathbf{C}$ with $\mathcal{V}$-relations on each homset such that:

$$\begin{array}{rcl} ; : \mathbf{C}(X,Y) \boxtimes_\oplus \mathbf{C}(Y,Z) & \to & \mathbf{C}(X,Z) \\ f \qquad\qquad g & \mapsto & f \,;\, g \\ d(f,f') \oplus d(g,g') & \sqsubseteq & d(f\,;\,g, f'\,;\,g') \end{array} \qquad \text{is a morphism in } \mathcal{V}\mathbf{Rel}$$

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$$\frac{f =_{\varepsilon} f' \qquad g =_{\delta} g'}{f\,;g =_{\varepsilon \oplus \delta} f'\,;g'}$$

Outline

Intuition

String diagrams are a syntax for processes and systems.

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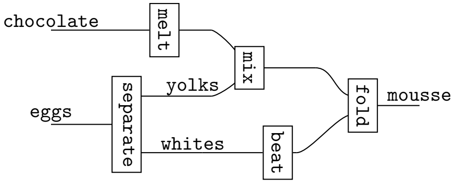

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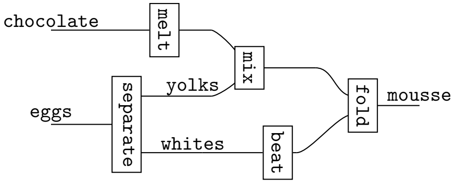

fold (,   )

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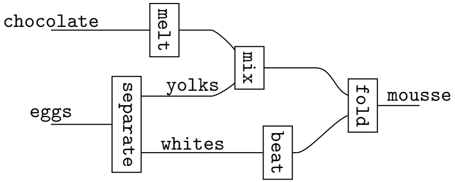

$\texttt{fold}\,(\texttt{mix}(\qquad\qquad , \qquad\qquad), \qquad\qquad)$

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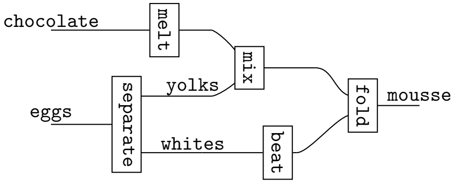

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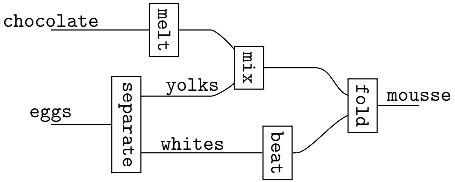

$\mathtt{fold}\,(\mathtt{mix}(\mathtt{melt}(\mathtt{chocolate}),\quad (\mathtt{separate}(\mathtt{eggs}))),\qquad\qquad\qquad\qquad)$

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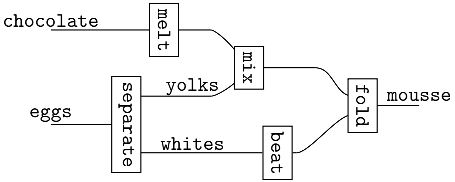

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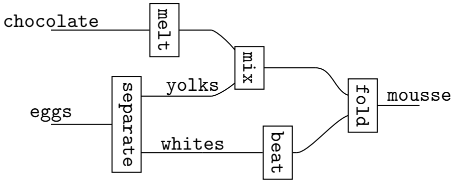

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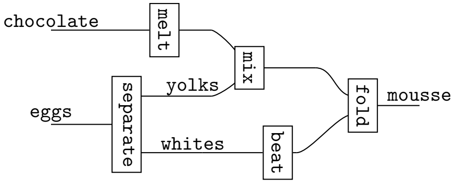

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$$(\mathtt{melt} \otimes \mathtt{separate}); (\mathtt{mix} \otimes \mathtt{beat}); \mathtt{fold}$$

Monoidal Terms

A **generator** is a symbol with an arity u and a coarity v in $\mathbb{N}$: $u \,-\!\boxed{f}\!-\, v$.
With a set of generators Σ, we inductively generate the set of monoidal terms $\mathcal{M}_\Sigma$.

$$\frac{u \,-\!\boxed{f}\!-\, v \in \Sigma}{u \,-\!\boxed{f}\!-\, v \in \mathcal{M}_\Sigma}\ \textsc{Gen} \qquad \frac{x, y \in \mathbb{N}}{\begin{smallmatrix} x \\ y \end{smallmatrix} \times \begin{smallmatrix} y \\ x \end{smallmatrix} \in \mathcal{M}_\Sigma}\ \textsc{Swap} \qquad \frac{x \in \mathbb{N}}{x \,\text{——}\, x \in \mathcal{M}_\Sigma}\ \textsc{Id} \qquad \frac{}{\cdots\cdots \in \mathcal{M}_\Sigma}\ \textsc{Unit}$$

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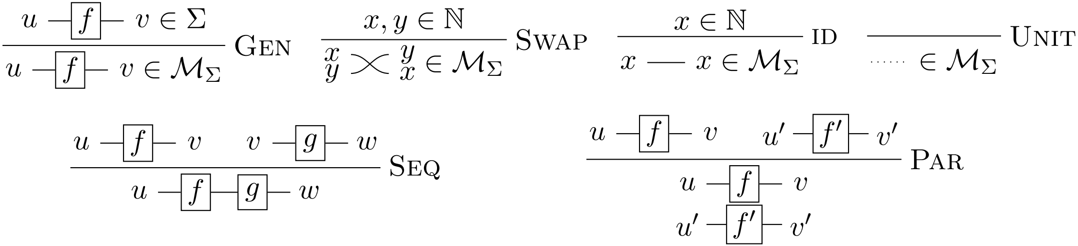

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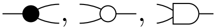
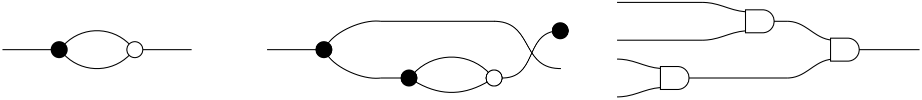
With generators .



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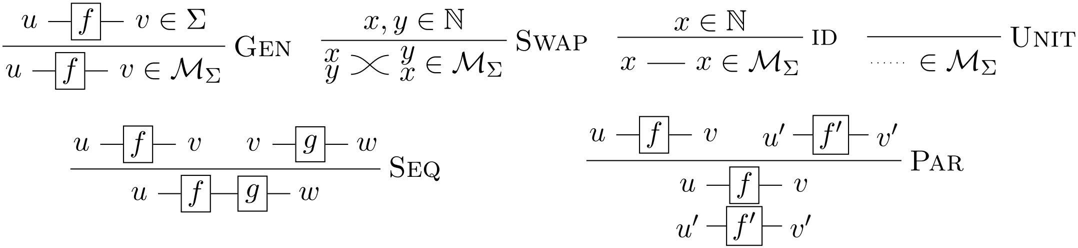

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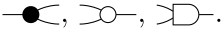
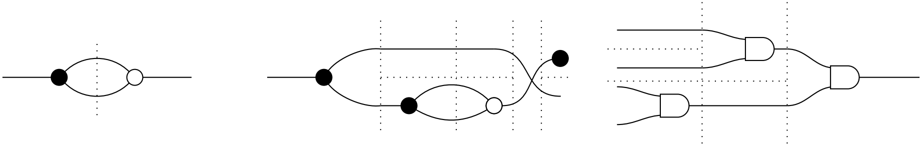
With generators 



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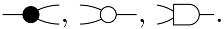
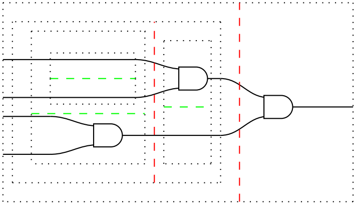
With generators .



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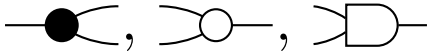
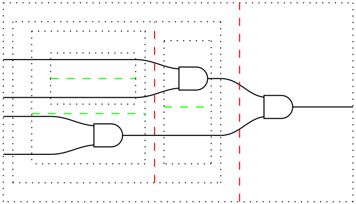
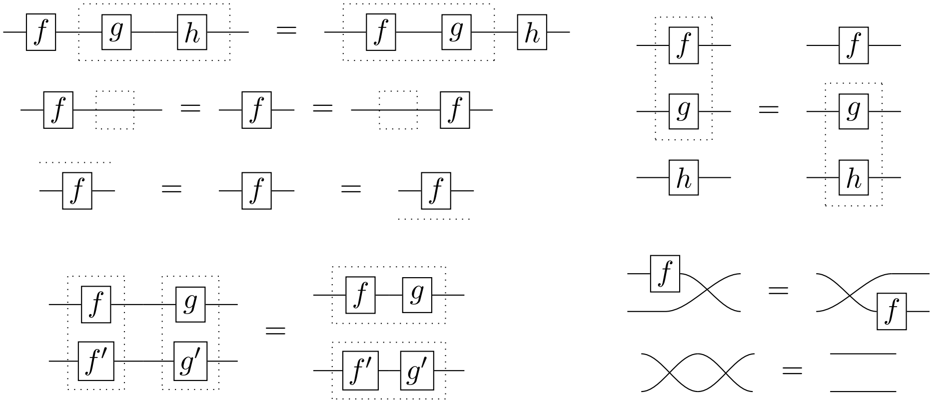
With generators .

 $= (((\mathrm{id} \otimes \mathrm{id}) \otimes \texttt{and})\ ;\ (\texttt{and} \otimes \mathrm{id}))\ ;\ \texttt{and}$

String Diagrams

String diagrams are monoidal terms considered modulo the following equations.

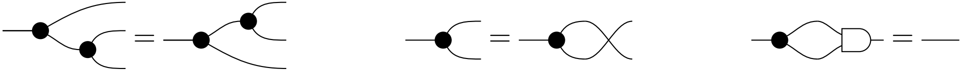


By [JS91], they are considered modulo continuous deformations.

[JS91] André Joyal and Ross Street. "The geometry of tensor calculus. I". In: *Adv. Math.* 88.1 (1991), pp. 55–112. ISSN: 0001-8708,1090-2082. DOI: 10.1016/0001-8708(91)90003-P. URL: https://doi.org/10.1016/0001-8708(91)90003-P

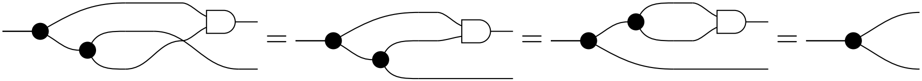
Equations

We can impose further equations between terms of the same type, e.g.,



Diagrammatic reasoning consists of:

- Treating string diagrams up to visually sound manipulations.
- Applying equational reasoning by replacing diagrams with other diagrams that are known to be equal.



Definition

Given a set of generators Σ and a set of equations E: define for any $u, v \in \mathbb{N}$,

$$\mathcal{S}_{\Sigma,E}(u,v) = \left\{ u \,-\!\boxed{f}\!-\, v \text{ built with } \Sigma \right\} /\text{diagrammatic reasoning with } E.$$

This freely generates the **syntactic category** $\mathcal{S}_{\Sigma,E}$.

Presentations

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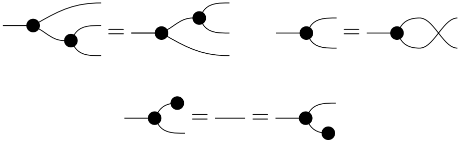
This freely generates the **syntactic category** $\mathcal{S}_{\Sigma,E}$.
A **presentation** for an SMC $\mathbf{C}$ is a pair (Σ, E) such that $\mathcal{S}_{\Sigma,E} \cong \mathbf{C}$.

$$\text{Morally, } \quad \frac{\text{string diagrams } u \;\text{—}\boxed{f}\text{—}\; v}{\overline{\text{morphisms } f : u \to v \in \mathbf{C}}}$$

Example 1: Finite Sets and Functions

Generators



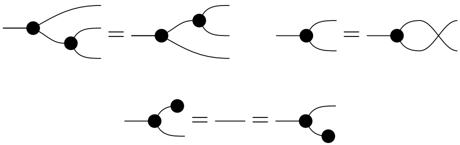
Equations

$\llbracket$  $\rrbracket : \{0, 1\} \to \{0\} = 0 \mapsto 0, 1 \mapsto 0$ $\qquad$ $\llbracket$  $\rrbracket : \{\} \to \{0\} = \_ \mapsto$

Semantics

Example 1: Finite Sets and Functions

Generators

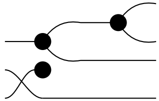


Equations

⟦⟧ $: \{0,1\} \to \{0\} = 0 \mapsto 0, 1 \mapsto 0$ ⟦⟧ $: \{\} \to \{0\} = \_ \mapsto$

Semantics

Example



$$\Longleftrightarrow \begin{cases} 0, 1, 2 & \mapsto 0 \\ 3 & \mapsto 1 \end{cases}$$

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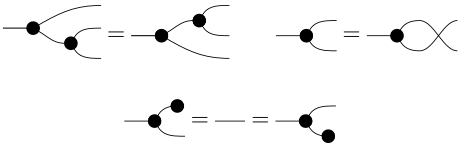
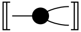


Equations

$[\![$  $]\!] : \{0,1\} \to \{0\} = 0 \mapsto 0, 1 \mapsto 0$ $\qquad$ $[\![$  $]\!] : \{\} \to \{0\} = \_ \mapsto$

Semantics

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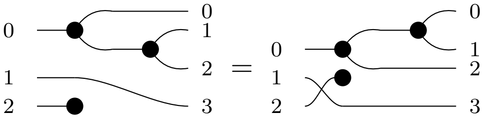
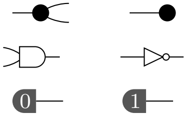


$$\iff \begin{cases} 0,1,2 & \mapsto 0 \\ 3 & \mapsto 1 \end{cases}$$

Example 2: Boolean Circuits



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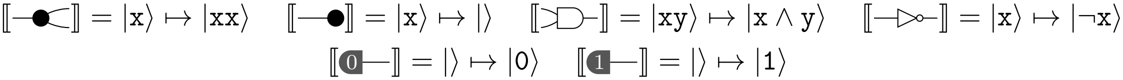

Semantics

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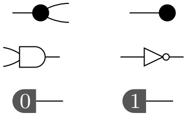

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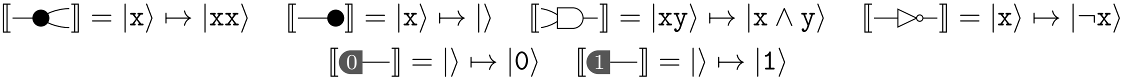

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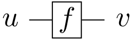

$$f : \mathbb{B}^u \to \mathbb{B}^v$$

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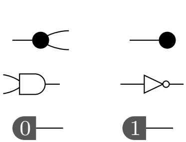

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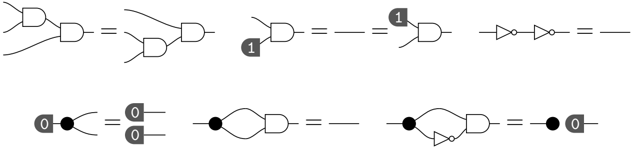
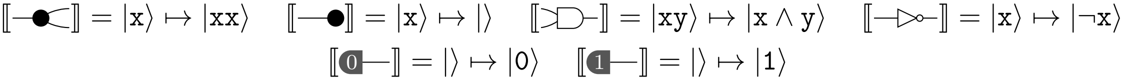


Some equations



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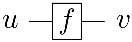

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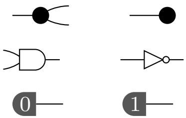

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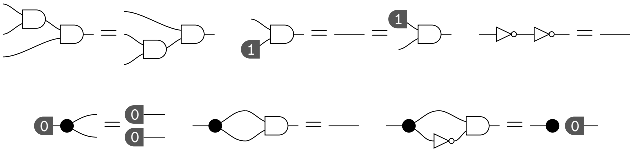

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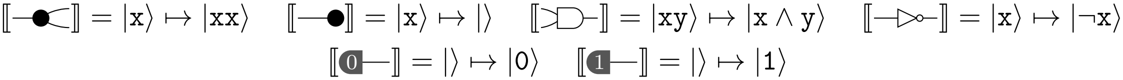

Semantics

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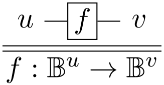

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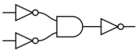

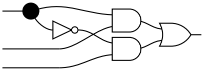

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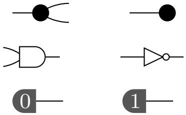

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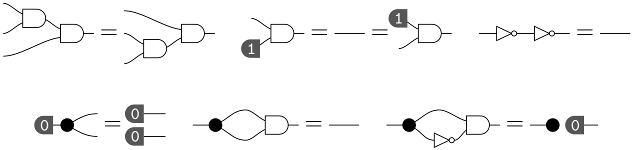
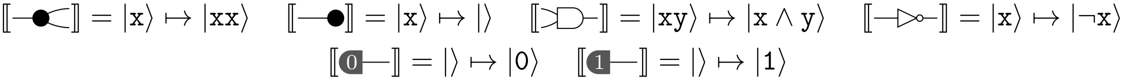


Some equations



Semantics

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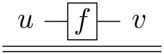

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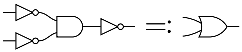

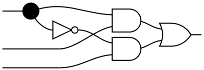

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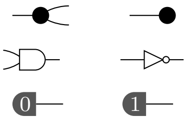

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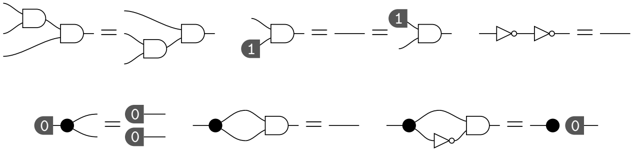

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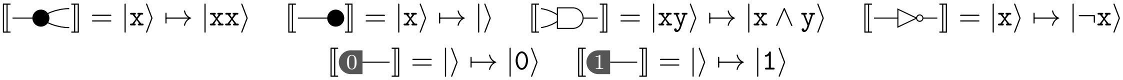

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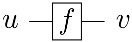

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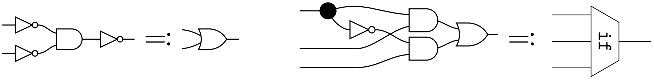

Example 3: Probabilistic Boolean Circuits

$$\mathsf{BoolCirc} +$$

$$p\!\!-\!\!-\ , \quad p \in (0,1)$$

Generators

$$[\![\, p\!\!-\!\!- \,]\!] = |\rangle \mapsto p|\mathsf{1}\rangle + (1-p)|\mathsf{0}\rangle$$

Semantics

string diagram $u -\!\!\boxed{f}\!\!- v$

probabilistic map $f : \mathbb{B}^u \to \mathbb{B}^v$

$2^v \times 2^u$ stochastic matrix

Example 3: Probabilistic Boolean Circuits

BoolCirc +

$\boxed{p}\!-\!-$, $p \in (0,1)$

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$[\![\, \boxed{p}\!-\!- \,]\!] = |\rangle \mapsto p|1\rangle + (1-p)|0\rangle$

Semantics

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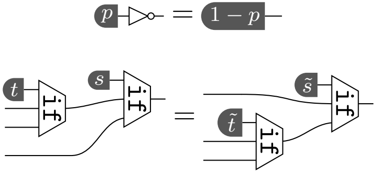

Some equations

$$\frac{\dfrac{\text{string diagram } u \;-\!\boxed{f}\!-\; v}{\text{probabilistic map } f : \mathbb{B}^u \to \mathbb{B}^v}}{2^v \times 2^u \text{ stochastic matrix}}$$

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Semantics

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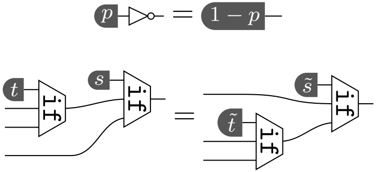
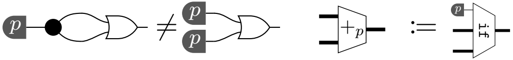


Some equations

Examples

string diagram u —$\boxed{f}$— v

$$\frac{\frac{\text{string diagram } u \text{ —}\boxed{f}\text{— } v}{\text{probabilistic map } f : \mathbb{B}^u \to \mathbb{B}^v}}{2^v \times 2^u \text{ stochastic matrix}}$$



Outline

String Diagrams and Quantitative Equations

Goal: String diagrams that form a $\mathcal{V}$**Rel**-enriched symmetric monoidal category.

$$\forall \varepsilon \sqsupseteq \varepsilon',\ -\boxed{f}- =_{\varepsilon} -\boxed{g}- \quad\Rightarrow\quad -\boxed{f}- =_{\varepsilon'} -\boxed{g}-$$

$$\left(\forall i\ -\boxed{f}- =_{\varepsilon_i} -\boxed{g}-\ \right) \Rightarrow\quad -\boxed{f}- =_{\sqcup_i \varepsilon_i} -\boxed{g}-$$

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$$-\boxed{f}- \ =_{\varepsilon} \ -\boxed{f'}- \quad , \quad -\boxed{g}- \ =_{\delta} \ -\boxed{g'}- \quad \Rightarrow \quad -\boxed{f}-\boxed{g}- =_{\varepsilon \oplus \delta} -\boxed{f'}-\boxed{g'}-$$

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$$\begin{gathered} \emptyset \Rightarrow f = f \qquad f = g \Rightarrow g = f \qquad f = g, g = h \Rightarrow f = h \\ f = f', g = g' \Rightarrow f \mathbin{;} g = f' \mathbin{;} g' \qquad f = f', g = g' \Rightarrow f \otimes g = f' \otimes g' \\ f = f', f =_{\varepsilon} g \Rightarrow f' =_{\varepsilon} g \qquad g = g', f =_{\varepsilon} g \Rightarrow f =_{\varepsilon} g'. \end{gathered}$$

Definition

Given Σ and a set of axioms E (with the shape $\Gamma \Rightarrow \varphi$), let $\vdash$ to be smallest consequence relation containing E and the implications in previous slide.

$$\text{Quotient } \mathcal{S}_{\Sigma,E}(u,v) \text{ by } u \;-\!\boxed{f}\!-\; v = u \;-\!\boxed{g}\!-\; v \Leftrightarrow \emptyset \vdash f = g$$

$$\text{Define } d(f,g) = \sup\{\varepsilon \mid \emptyset \vdash f =_{\varepsilon} g\}$$

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This freely generates the **syntactic $\mathcal{V}$Rel-category** $\mathcal{S}_{\Sigma,E}$.
A **presentation** for an $\mathcal{V}$**Rel**-SMC $\mathbf{C}$ is a pair (Σ, E) such that $\mathcal{S}_{\Sigma,E} \cong \mathbf{C}$.

Case Study: Kullback–Leibler Divergence

Let $\mathcal{D}X$ denote the set of *probability distributions* over a set X, namely,

$$\mathcal{D}X \coloneqq \{\varphi : X \to [0,1] \mid \textstyle\sum_{x\in X} \varphi(x) = 1\}.$$

KL divergence [KL51] is a $[0,\infty]_+$-relation on $\mathcal{D}X$.

$$\mathsf{kl} : \mathcal{D}X \times \mathcal{D}X \to [0,\infty]_+$$

$$\mathsf{kl}(\varphi,\psi) \coloneqq \sum_{x\in X} \varphi(x) \log \frac{\varphi(x)}{\psi(x)}, \text{ with } \log \frac{x}{0} = \infty \text{ and } 0 \cdot \infty = 0$$

[KL51] S. Kullback and R. A. Leibler. "On information and sufficiency". In: *Ann. Math. Statistics* 22 (1951), pp. 79–86. ISSN: 0003-4851. DOI: 10.1214/aoms/1177729694. URL: https://doi.org/10.1214/aoms/1177729694

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For probabilistic maps $f, g : \mathbb{B}^u \to \mathbb{B}^v$,

$$\overline{\mathsf{kl}}(f,g) := \max \mathsf{kl}(f|\mathtt{x}_1,\ldots,\mathtt{x}_u\rangle, g|\mathtt{x}_1,\ldots,\mathtt{x}_u\rangle)$$

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Case Study: Kullback–Leibler Divergence

$$\mathsf{kl}(\varphi, \psi) = 0 \Leftrightarrow \varphi = \psi$$

$$\overline{\mathsf{kl}}(f \,;\, g, f' \,;\, g') \leq \overline{\mathsf{kl}}(f, f') + \overline{\mathsf{kl}}(g, g') \qquad \overline{\mathsf{kl}}(f \otimes g, f' \otimes g') \leq \overline{\mathsf{kl}}(f, f') + \overline{\mathsf{kl}}(g, g')$$

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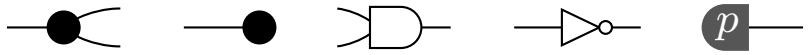
if $\varphi = p|\mathsf{1}\rangle\varphi_{\mathsf{1}} + (1-p)|\mathsf{0}\rangle\varphi_{\mathsf{0}}$ and $\psi = q|\mathsf{1}\rangle\varphi_{\mathsf{1}} + (1-q)|\mathsf{0}\rangle\psi_{\mathsf{0}}$, then

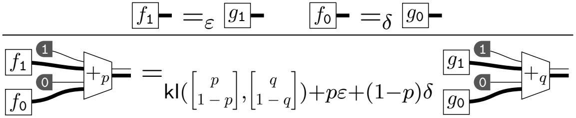
$$\mathsf{kl}(\varphi, \psi) = \mathsf{kl}\left(\begin{bmatrix} p \\ 1-p \end{bmatrix}, \begin{bmatrix} q \\ 1-q \end{bmatrix}\right) + p\,\mathsf{kl}(\varphi_{\mathsf{1}}, \psi_{\mathsf{1}}) + (1-p)\mathsf{kl}(\varphi_{\mathsf{0}}, \psi_{\mathsf{0}})$$

Theorem

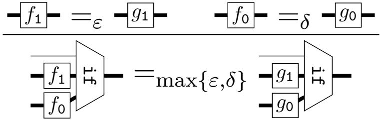
The category of stochastic matrices enriched with KL divergence is presented by



$$\frac{f = g \in \mathsf{ProbCirc}}{f = g} \qquad \frac{}{f =_0 f}$$



Chain$_\otimes$



If$_{\max}$

Proof

We need to prove that

$$\emptyset \vdash f =_{\varepsilon} g \text{ if and only if } \overline{\mathsf{kl}}(f, g) \leq \varepsilon$$

($\Rightarrow$) is by the properties of KL divergence.
($\Leftarrow$) is done with two inductions.

Proof

Lemma

For any $f, g : 0 \to m$, *if* $\varepsilon = \overline{\mathsf{kl}}(\llbracket f \rrbracket, \llbracket g \rrbracket)$, *then* $\emptyset \vdash f =_\varepsilon g$.

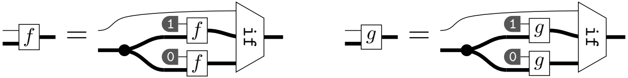
By induction on m, base case is handled by $f =_0 f$. Let $f, g : 0 \to m + 1$, then

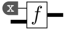
Then, we use the induction hypothesis on $f_{\mathtt{x}}$ and $g_{\mathtt{x}}$, and $\textsc{Chain}_\otimes$

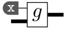
Proof

Lemma
For any $f, g : n \to m$, if $\varepsilon = \overline{\mathsf{kl}}(\llbracket f \rrbracket, \llbracket g \rrbracket)$, then $\emptyset \vdash f =_\varepsilon g$.

By induction on n. Base case is handled by previous lemma. Let $f, g : n + 1 \to m$, then



Then, we use the induction hypothesis on 

and 

, and $\textsc{If}_{\max}$.

Future Work

More Examples

- *Already done*: KL divergence, Rényi divergence, total variation
- *Looking into*: quantum relative entropy, diamond distance
- *Dream*: distance for evaluating probabilistic learning models

Quantitative Program Logics

- Diagrammatic Hoare logics with tape diagrams in [BDGDL25]
- Quantitative Hoare logics in many papers.
- *Dream*: Quantitative diagrammatic Hoare logics

Extend the Framework

- *Fibered/Doubled*: to account for distances between in/outputs (Kantorovich)
- *Graded*: for an even finer comparison.

[BDGDL25] Filippo Bonchi, Alessandro Di Giorgio, and Elena Di Lavore. "A diagrammatic algebra for program logics". In: *Foundations of software science and computation structures*. Vol. 15691. Lecture Notes in Comput. Sci. Springer, Cham, 2025, pp. 308–330. ISBN: 978-3-031-90896-5; 978-3-031-90897-2. DOI: 10.1007/978-3-031-90897-2_15. URL: https://doi.org/10.1007/978-3-031-90897-2_15

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